

Which Features are Actually Effective?

A common flaw prevails across existing CAM-based methods: they often fail to fully capture the features that the model truly relies on, leading to explanations that do not faithfully reflect the model's actual inference process.

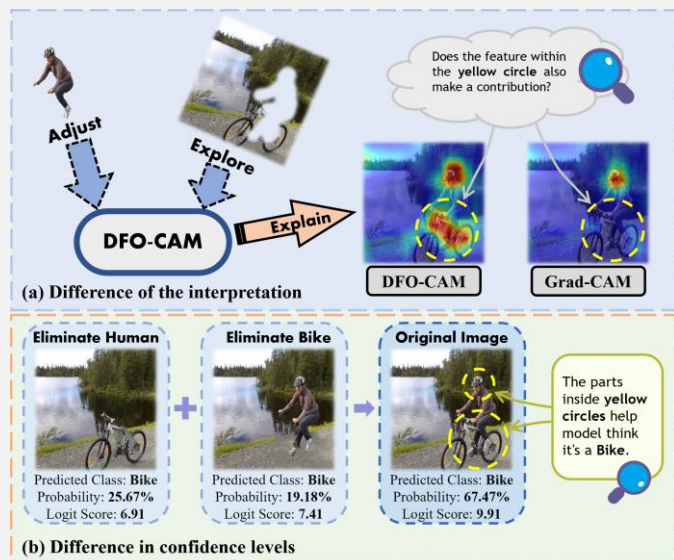


Fig.1 Illustration of our proposed method and current problem.

Specifically, we define this issue as the **faithfulness** problem of explanation methods.

The Definition of Faithfulness

We define **faithfulness** as the extent to which an attribution map completely and correctly reflects the evidence used by the model to produce a prediction.

Additional proofs, experimental results, methodological details, and source code are available at <https://github.com/CQUPT-JX-Mi-Lab/2026TIP-BFO-CAM>.

Dual-Flow Optimization-CAM

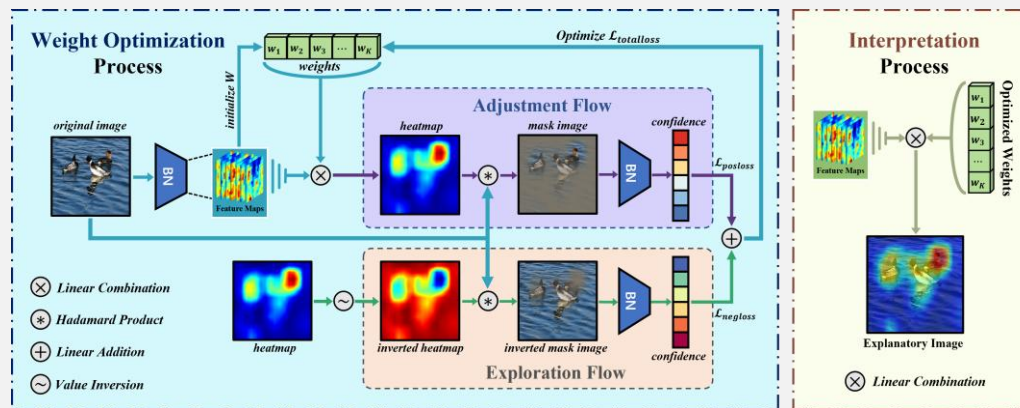


Fig.2 The flowchart of the Dual-Flow Optimization-CAM method.

- Weakly activated regions in the initial explanation heatmap still contain **latent decision features** with positive contributions, and **value inversion** uncovers critical evidence omitted by conventional CAM methods.
- Feature maps are fused via **optimizable channel weights** to generate a **forward mask**, whose inversion yields an **inverted mask** corresponding to explicit features and unexplored latent features respectively.
- A **dual-flow joint loss function** is constructed to jointly optimize the classification confidence of both masked inputs, iteratively aligning explanations with the model's **actual decision-making logic**.

Qualitative Experiment

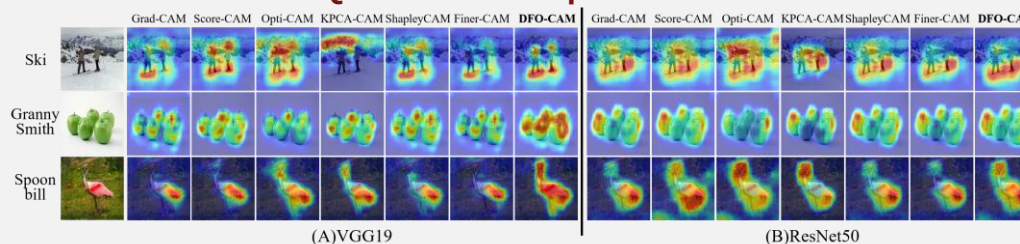


Fig.3 Qualitative comparison of CAM-based explanation methods on VGG-19 and ResNet-50.

Quantitative Experiment

Datasets	PASCAL VOC 2012 [32]						CUB-200 2011 [31]					
	ResNet-50 (77.60%)	AD _l	AI _l	AG _l	VGG-19 (75.42%)	AD _l	AI _l	AG _l	ResNet-50 (86.31%)	AD _l	AI _l	AG _l
Grad-CAM [20]	10.53	50.8	31.43	26.69	30.8	15.01	27.04	24.2	8.18	69.61	16.1	3.03
Score-CAM [23]	8.11	54.1	35.22	21.73	33.5	18.00	20.52	28.3	9.19	40.68	31.3	9.78
BI-CAM [27]	8.43	54.8	33.71	22.44	33.4	17.56	19.05	30.2	9.34	30.99	38.4	12.82
Opti-CAM [16]	17.00	38.2	19.93	28.50	26.2	11.07	19.97	27.8	7.89	40.49	28.4	9.14
KPCA-CAM [25]	36.94	28.3	14.41	40.37	20.0	8.36	40.41	17.1	4.90	45.36	26.9	9.27
Finer-CAM [28]	9.80	52.9	32.84	26.74	31.1	14.98	28.95	22.7	7.55	55.25	21.9	6.04
Shapley-CAM [18]	8.63	53.0	34.18	26.16	31.2	15.99	21.51	28.5	9.04	46.43	27.9	8.00
DFO-CAM (Ours)	8.06	54.3	34.50	22.42	34.1	18.03	17.50	32.0	10.33	31.14	39.0	14.02

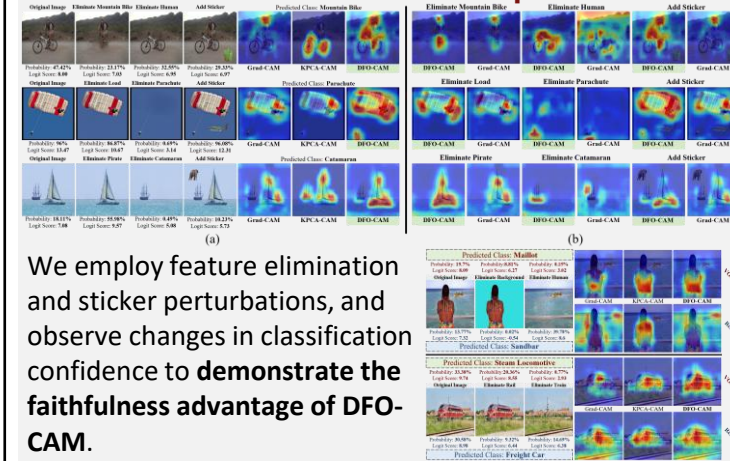
Datasets	ImageNet-1K 2012 [1]						CIFAR-10 [33]					
	ResNet-50 (76.15%)	AD _l	AI _l	AG _l	VGG-19 (72.39%)	AD _l	AI _l	AG _l	ResNet-50 (83.87%)	AD _l	AI _l	AG _l
Grad-CAM (2017)	31.73	25.8	10.35	58.90	14.2	4.41	69.57	8.8	3.51	72.46	9.9	1.53
Score-CAM (2020)	31.77	25.1	9.63	53.80	17.2	5.45	57.73	13.6	5.52	55.31	22.1	6.26
BI-CAM (2023)	26.07	28.6	11.31	51.51	16.8	5.37	58.49	13.7	5.47	54.51	17.0	4.16
Opti-CAM (2024)	33.35	22.5	8.07	52.67	14.4	5.18	61.79	10.6	3.97	64.58	16.4	3.45
KPCA-CAM (2024)	63.84	10.3	3.24	69.24	9.5	3.04	72.87	7.9	2.66	66.97	15.2	3.21
Finer-CAM (2025)	35.92	23.1	9.44	62.20	13.0	4.12	71.03	8.4	3.04	72.24	10.7	1.62
Shapley-CAM (2025)	31.64	26.3	10.32	53.67	17.1	5.70	67.40	9.9	3.87	73.01	9.8	1.55
DFO-CAM (Ours)	29.62	28.7	11.44	47.79	18.1	6.22	59.98	13.9	5.72	60.14	17.8	4.47

Tab.1 Faithfulness evaluation using AD/AI/AG on VGG-19 and ResNet-50 across four datasets.

Networks	GoogLeNet (62.44%)			EfficientNet-B4 (77.52%)			ConvNeXt-Tiny (81.87%)			ViT-B/16 (81.10%)		
	AD _l	AI _l	AG _l	AD _l	AI _l	AG _l	AD _l	AI _l	AG _l	AD _l	AI _l	AG _l
Grad-CAM	73.33	9.3	1.31	64.92	19.6	4.70	62.64	16.2	4.82	42.33	13.0	3.30
Score-CAM	70.96	10.4	1.79	91.70	4.8	0.62	49.25	13.6	5.99	48.59	12.8	3.00
BI-CAM	68.89	11.2	1.92	86.47	6.9	1.65	30.83	37.4	8.36	44.88	13.9	4.01
Opti-CAM	75.83	7.2	0.95	91.83	3.9	0.29	83.60	6.6	1.48	52.57	12.0	1.99
KPCA-CAM	84.00	5.7	0.74	54.91	23.2	5.99	31.33	25.8	6.49	71.20	9.4	1.95
Finer-CAM	75.25	8.6	1.21	65.09	17.7	5.24	52.15	19.8	6.01	50.07	11.2	2.81
Shapley-CAM	73.44	9.0	1.23	28.23	38.7	10.42	62.40	16.3	4.76	/	/	/
DFO-CAM (Ours)	67.37	11.3	2.06	43.27	31.9	9.60	30.37	32.7	9.18	41.42	16.8	3.87

Tab.2 Faithfulness evaluation using AD/AI/AG across additional models on ImageNet-1K.

Faithfulness Ablation Experiment



We employ feature elimination and sticker perturbations, and observe changes in classification confidence to **demonstrate the faithfulness advantage of DFO-CAM.**